

SAMPLE - NOT ISSUED. No Engagement Terms are incorporated and no commitment of any kind is made. Issued reports carry the governing terms version here.

Order identity

Field	Value
Client	Acme Labs
Order id	LQI-2026-Troesch-15-2026-08-29-T2
Reference	LQI-2026-Troesch-15
Problem	A plasma-confinement model.
Stated equation & data	$y'' = 15 \cdot \sinh(15 \cdot y)$, $y(0) = 0$, $y(1) = 1$
Ordered (date in)	2026-08-29
Delivered (date out)	2026-08-29
Service tier	Tier 2: Certified Standard

CERTIFICATE - Tier 2 (Certified Standard)

- Class: ODE boundary-value problem | Engine: SIRPY native — piecewise power series (35 pieces) | Generated by SIRPY 0.59.15
- FINDING: solved+verified
- max residual (measured): 1.588e-03 at $x = 0.999995$
- artifact coverage: $[0, 1]$; pieces solved 35, delivered 35
- artifact: LQI-2026-Troesch-15-T2.artifact.json
- sha256: c751490c d3108fa3 01eaaffb bbaff6c4 bc5b7849 4a0d801d 90f0a185 3cc35704
- reproduction: the headline numbers above are RECOMPUTABLE from this artifact and the stated equation using only Python and sha256sum -- substitute the delivered coefficients into the equation and compare. This is reproduction of our measurements, not agreement with a third-party solver: an independent solver may be compared against the delivered values, and any disagreement is a finding to report, not a covered reproducibility failure. Scope: the artifact coverage above.

Claim 1 — Constraint satisfaction (measured)

Equation residual by exact coefficient differentiation at dense off-node points: max 1.588e-03 (relative 2.984e-10).

Boundary/initial data residual: 5.845e-10.

Checked: the equation (residual by exact coefficient differentiation); boundary data.

Claim 2 — Seam continuity (measured)

Max one-sided jump across 34 interior node(s): 2.349e-12 - each side's own coefficients evaluated exactly at the node. The per-node figure in this package plots this same measurement, annotated with this same value.

Claim 3 — Measured tail decay and trust radius

y: coefficient tail/head ratio per piece - median $8.1\text{e}+22$, worst $6.3\text{e}+44$ over 35 piece(s); a MEASURED decay indicator, not a continuous-interval remainder bound.

See the footnote "How to read the tail decay ratio" at the end of this section.

Claim 4 — Independent forward check

exact energy-quadrature relation: $5.343\text{e}-11$.

Characteristic values (from delivered data)

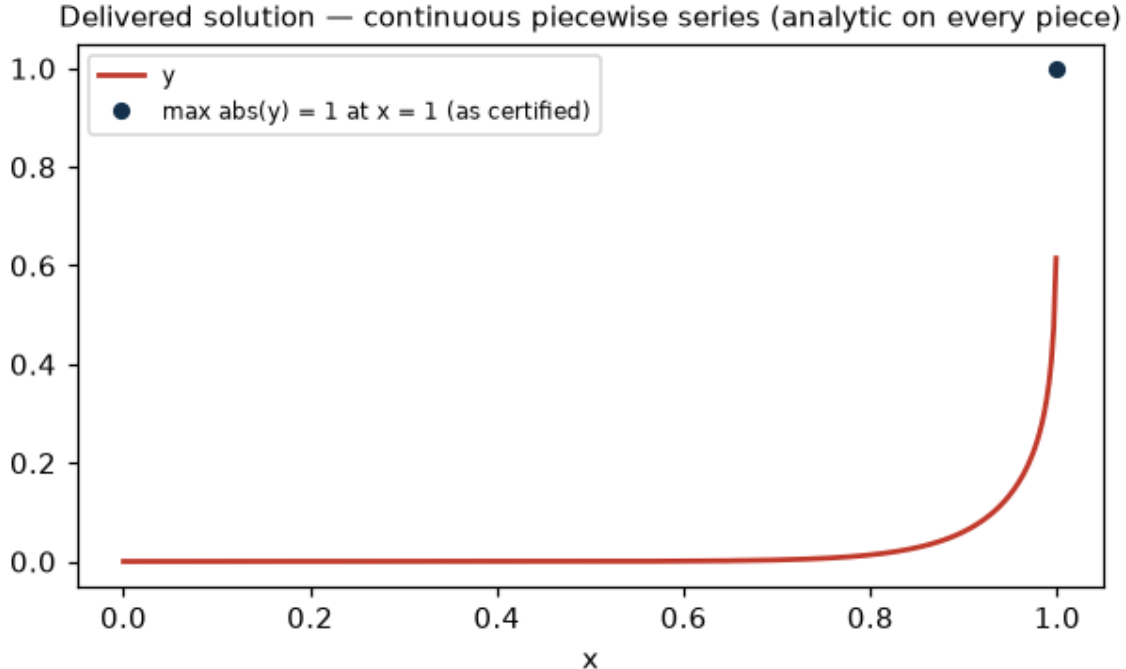
$\max \text{abs}(y) = 1$ at $x = 1$ [computed from the delivered series representation]

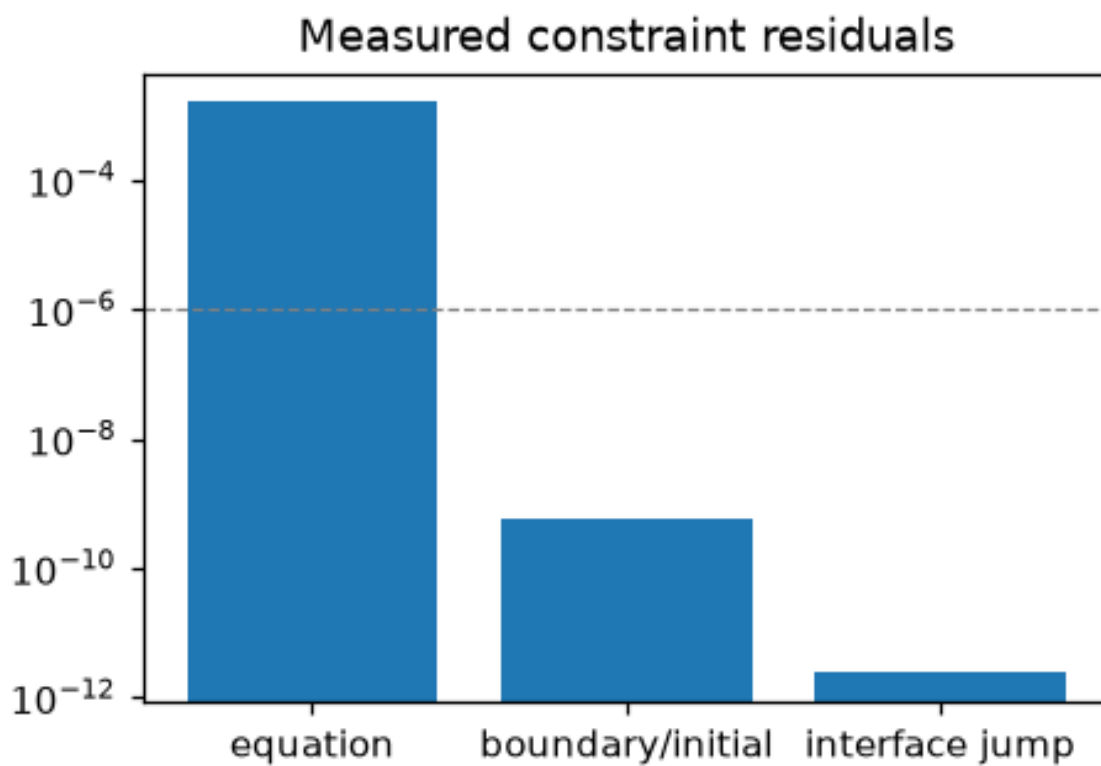
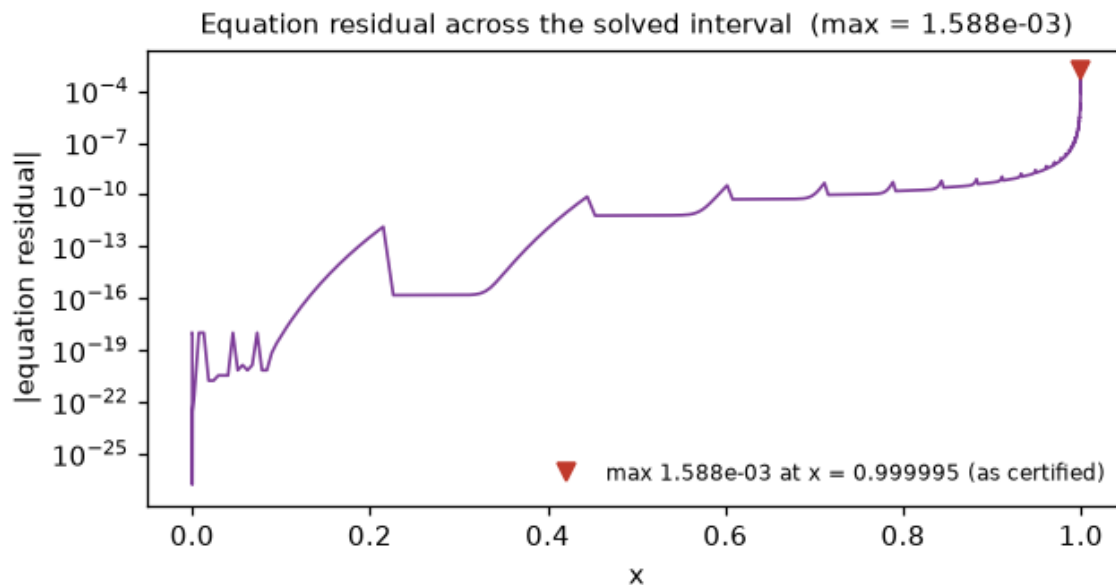
Notes

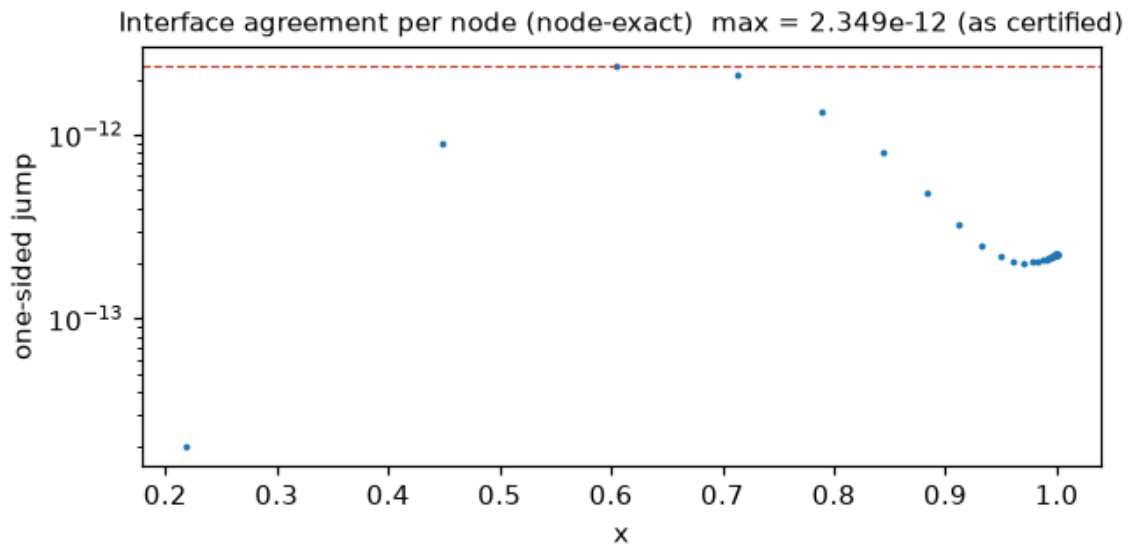
How to read the tail decay ratio. It divides a piece's tail coefficients by its leading ones. Where the solution is very small, the leading coefficients are near zero, so the ratio grows large even though the piece itself is accurate - a large ratio means we divided by something tiny, NOT that the error is large. The authoritative accuracy figures are the measured residuals in Claim 1; if a piece is genuinely stretched, the certificate says so and names the lever (partition / iterations).

Scope: this tier certifies the four claims above only. Diagnostics, localization, and analytics are Tier-3 deliverables.

Delivered figures







Issued by LQI. The order manifest (ORDER_AND_DELIVERY.txt) is included in this delivery package alongside this document and the artifact; the terms of service applicable to this order are those supplied with your order confirmation and named by version in the delivery manifest.